SC-6 QUEST SCOREBOARD & +10 TOKEN KIT

Daily Multi-Method Mastery Scoreboard (Sockets 1–10, 100 & ? Badge)

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · **9·Quest** · 10·Rack · 11·SC-32

CARD 10 / 10 • QUEST-01

MATHEMATICAL REFERENCE

1. WHAT IS THIS MANIPULATIVE?

ONE PROBLEM. MULTIPLE WAYS. SCORE YOUR MASTERY.

Confidence grows by solving in multiple ways.
Reward each approach with tactile score tokens:
1 Block = +10 (1st Way) • 2 Blocks = +20 (2nd Way)
Stacking a 2nd token rewards a 2nd solution method!

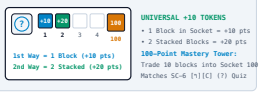
UNIVERSAL +10 QUARTER-BLOCK DATUM

- Token: Universal +10 Block = 10.25×10.25×7.0mm
- 1st Way: Seat 1 token in socket 1–10 (+10 pts)
- 2nd Way: Stack 2nd token on top (+20 pts total)
- Mastery: Trade 10 tokens into [100] socket (70mm)

2-PIECE SNAPFINITY SCOREBOARD LID (NO INNER TRAY)

Storage bin holds all 60 tokens under Snapfinity lid.
Use 12 lid sockets (1–10, 100, ?) for daily practice!

LID SOCKETS 1–10 & 100: 1 BLOCK = +10, 2 STACKED = +20



UNIVERSAL +10 TOKENS

- 1 Block in Socket = +10 pts
- 2 Stacked Blocks = +20 pts

100-Point Mastery Tower:
Trade 10 blocks into Socket 100
Matches SC-6 [↑][C] (?) Quiz

1st Way = 1 Block (+10 pts)
2nd Way = 2 Stacked (+20 pts)

2. KIT PARTS INVENTORY (62 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
1st-Way +10 Token	30x	10.25x10.25x7.0mm	Red+Gold Silk
2nd-Way +10 Token	30x	10.25x10.25x7.0mm	Glow Yellow
3x1x7U Open Bin	1x	125.5x42.0x56.4mm	Technical White
Scoreboard Lid	1x	Sockets 1–10, 100	Technical White

TOTAL: 60 Universal +10 Tokens + 7U Bin + Lid = 62 Parts

DUAL-TONE +10 TOKEN PALETTE

- Interchangeable Tokens: All 60 quarter-blocks are +10 pts.
- Red+Gold Silk (30x): Seat for 1st Way solution (+10 pts).
- Glow-in-the-Dark Yellow (30x): Stack on top for 2nd Way (+20 pts)!
- High Visibility: Easy to tally scores at a glance.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **5-SECOND OPEN-BIN PACK-UP**

No inner sorting tray needed! Sweep all 60 universal +10 tokens directly into the open 3x1x7U Gridfinity bin, lay this reference card on top, and snap the Scoreboard Lid shut for silent, zero-rattle classroom storage.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

PÓLYA'S MULTIPLE-PATH PRINCIPLE
Solving one problem in two independent ways builds deeper mastery than solving ten by rote.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 10 OF 10 • DOMAIN: MULTI-METHOD MASTERY & DAILY FLUENCY • 60x +10 QUARTER-BLOCK TOKENS • SOCKETS 1–10 & 100

DAILY PROTOCOL: 10 QUESTIONS, 2 WAYS & THE 100 MASTERY TOWER

How to Use Your SC-6 Quest Scoreboard with Daily Practice & Left-Shift [C] (?) Tutorial Mode

Grades 3–8 Daily Math Fluency • Hands-On Classroom & Home STEM Studio Series

CARD 10 / 10 • QUEST-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/quest-scoreboard/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.PRACTICE.MP1 (Persevere & Verify)
- CCSS.MATH.PRACTICE.MP5 (Use Strategic Tools)
- SC-6 Built-In Scoring (+10 / +20 / 100 Mastery)

Grades 3–8 Daily Math Fluency

💡 **MATHEMATICAL INQUIRY**

Can you verify your textbook calculation using physical blocks or the SC-6 RPN stack to earn a second +10 token?

① STEP 1: SOLVE 1ST WAY (+10 TOKEN)

Place the Scoreboard Lid on your desk beside your worksheet.
When you solve Question 1 using your first method (Textbook, Blocks, or SC-6), seat one +10 token into Socket [1].
Work through Sockets [1] to [10] at your own pace!

② STEP 2: TRY A 2ND WAY (+20 STACK)

Pick any question and check it a SECOND way:
- Built it with blocks? Check with SC-6 RPN keystrokes!
- Solved by hand? Prove it with Prime or Fraction blocks!

Notice: Stack a 2nd +10 token right on top (= +20 pts)!

3–4 STEPS 3 & 4: BUILD THE [100] MASTERY TOWER

3. TRADE 10 TOKENS FOR 100: Every 10 tokens (100 points), stack them into a 10-block tower in Socket [100]!
4. LAUNCH SC-6 QUEST MODE: Press Left-Shift [f] then [C] (?) matching the Question Mark in a Circle socket.
Calculator Verification: Press Question Mark in a Circle Tutorial Icon ([f][7]) and run 8 [ENTER] 2 [+] 5 [+] [m] → 15.

⑤ CORE PRINCIPLE: WHY TWO WAYS BEATS ROTE DRILL

Students who think they are 'bad at math' gain immediate confidence when physical blocks, textbook steps, and the SC-6 calculator all land on the exact same answer. Every 2-block stack on your scoreboard is physical proof that mathematics is consistent and discoverable!

★ EXTENSION INVESTIGATION: EARN A 200-POINT DOUBLE MASTERY DAY

Solve all 10 daily problems using your 1st Way (100 pts), then verify 10 problems using a 2nd Way to stack 20 tokens (200 pts total)!
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

QUEST-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 62 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California